

into the frame when there were a number of methods, and algorithms existed for the same task then the question arose that which one is more efficient and the process of determining the efficient way is known as query optimization.

There are many benefits of query optimization:

- It reduces the time and space complexity.
- More queries can be performed as due to optimization every query comparatively takes less time.
- User satisfaction as it will provide output fast

24) What do you mean by durability in DBMS?

Once the DBMS informs the user that a transaction has completed successfully, its effect should persist even if the system crashes before all its changes are reflected on disk. This property is called durability. Durability ensures that once the transaction is committed into the database, it will be stored in the non-volatile memory and after that system failure cannot affect that data anymore.

25) What is normalization?

Normalization is a process of analysing the given relation schemas according to their functional dependencies. It is used to minimize redundancy and also used to minimize insertion, deletion and update distractions. Normalization is considered as an essential process as it is used to avoid data redundancy, insertion anomaly, updation anomaly, deletion anomaly.

There most commonly used normal forms are:

- First Normal Form(1NF)
 - Second Normal Form(2NF)
 - Third Normal Form(3NF)
 - Boyce & Codd Normal Form(BCNF)
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